

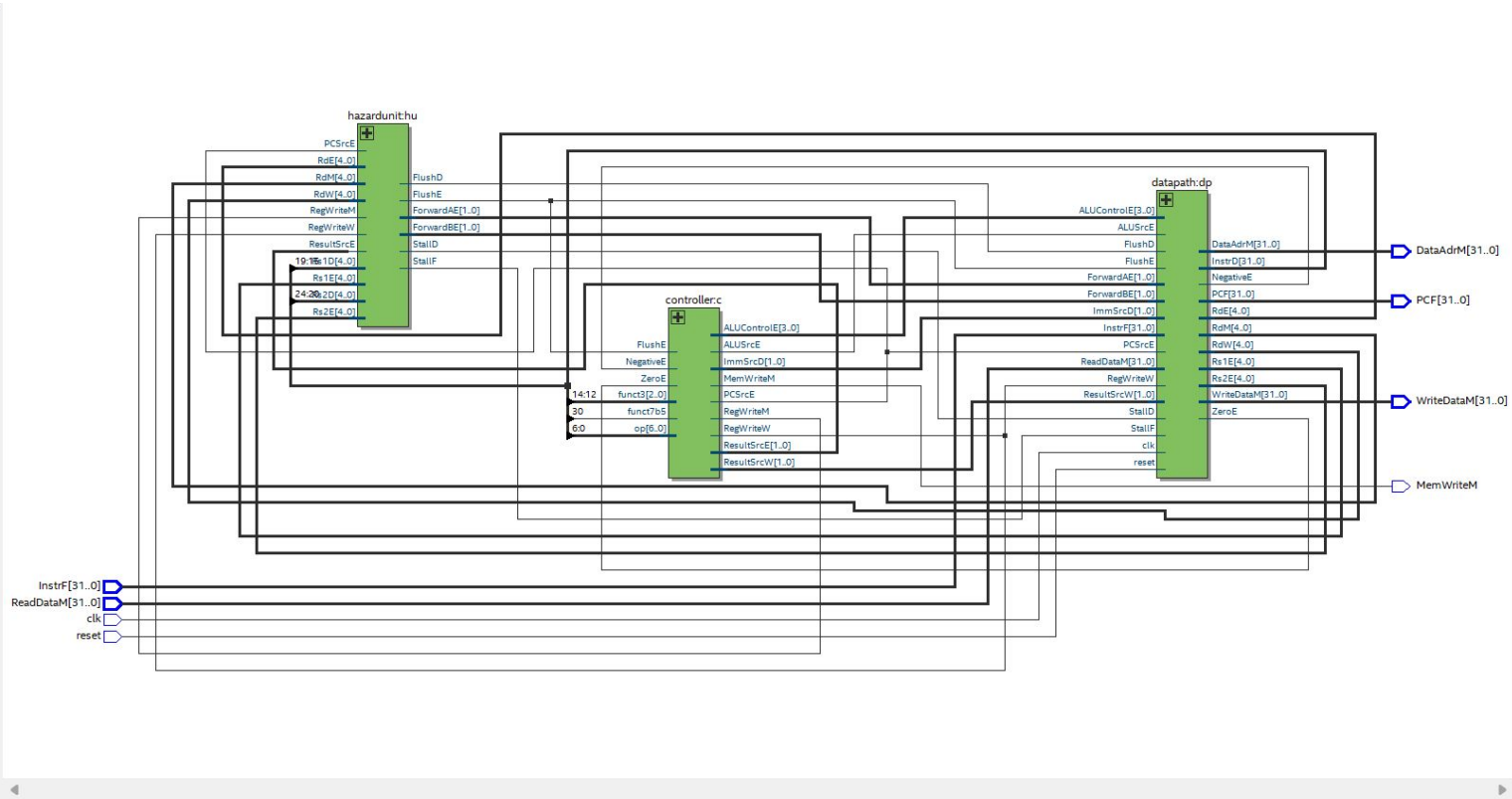
Main Decoder Truth Table

Instruction	Opcode	RegWrite	ImmSrc	ALUSrcA	ALUSrcB	MemWrite	ResultSrc	Branch	ALUOp	Jump
lw	0000011	1	000	0	1	0	01	0	00	0
sw	0100011	0	001	0	1	1	XX	0	00	0
R-type	0110011	1	0XX	0	0	0	00	0	10	0
beq	1100011	0	010	0	0	0	XX	1	01	0
I-Type ALU	0010011	1	000	0	1	0	00	0	10	0
jal	1101111	1	011	X	X	0	10	0	XX	1
lui	0110111	1	100	X	X	0	0	0	10	0

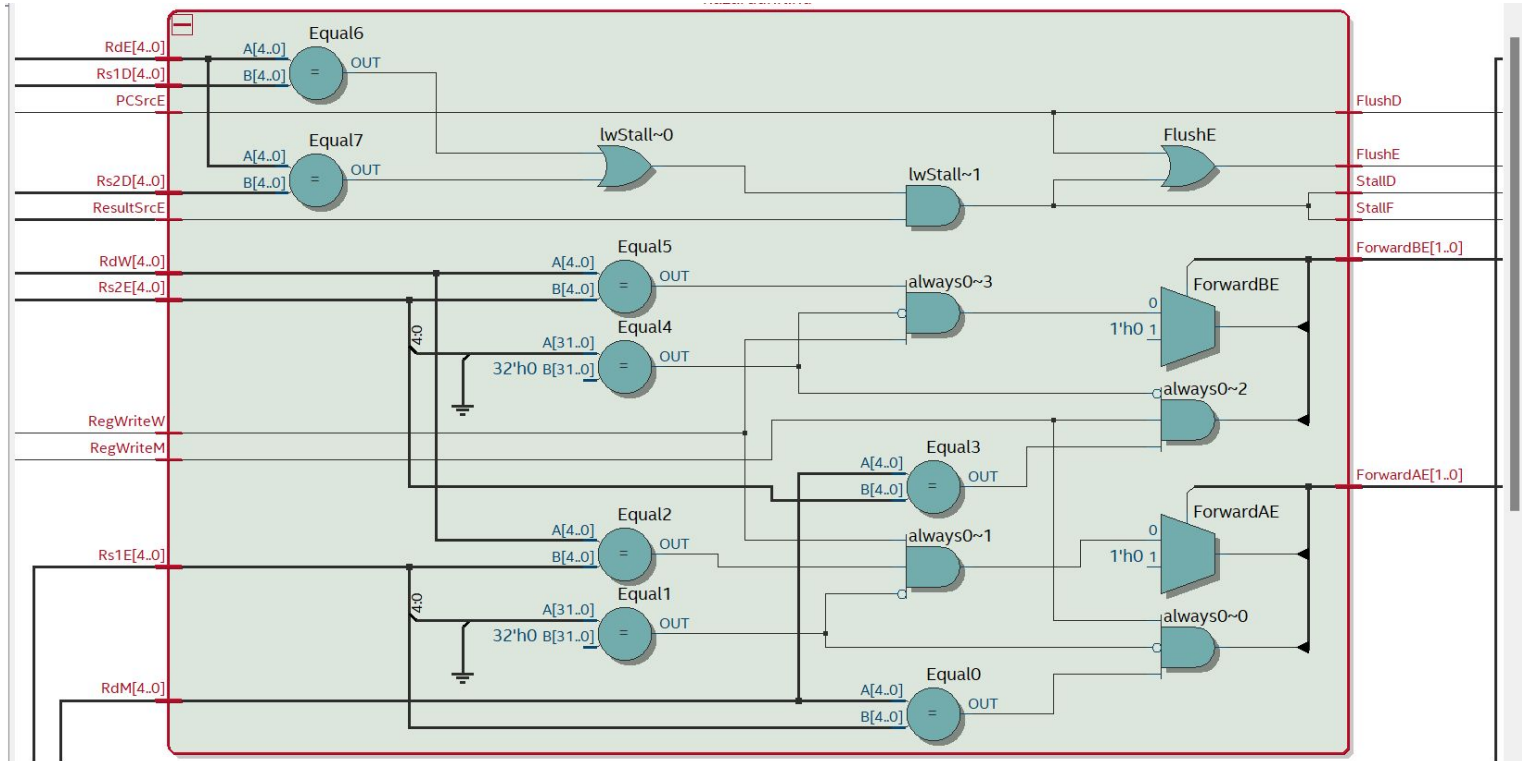
ALU Decoder Truth Table

<i>ALUOp</i> _{1:0}	<i>funct3</i> _{2:0}	<i>{op5, funct7}</i> ₅	<i>ALUControl</i> _{2:0}	Operation
00	x	x	000	Add
01	x	x	001	Subtract
10	000	00, 01, 10	000	Add
	000	11	001	Subtract
	010	x	101	SLT
	110	x	011	OR
	111	x	010	AND

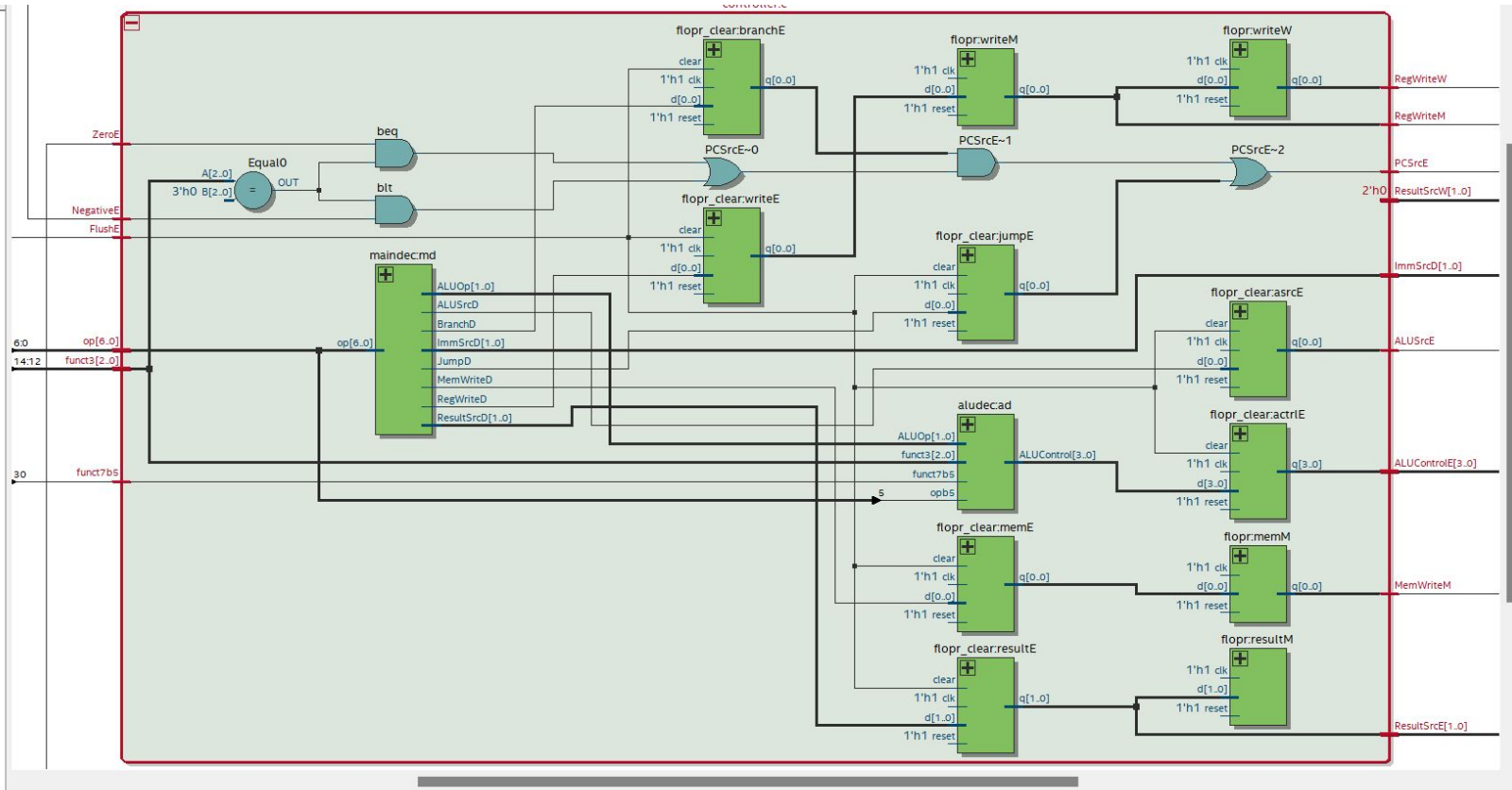
RTL Schematic Top View



RTL Schematic Hazard Unit



RTL Schematic Controller



RTL Schematic DataPath

